DESIGN SYSTEMS

# TECHNICAL DATA SHEET DS-POLTEC W A+B

## WATER-BASED TRANSPARENT POLYURETHANE FINISH – GLOSSY, SATIN AND MATT

## DESCRIPTION

DS-POLTEC W A+B is a two-component, water-based, transparent polyurethane enamel, consisting of a polyurethane dispersion cross-linked with a hydrophilic aliphatic polyisocyanate, available in glossy, satin and matt versions. Used as a finish for resin coatings, it makes it possible to create easy-to-clean surfaces. It adheres easily to numerous substrates and its characteristics (resistance to mechanical stress, abrasion, solvents, ethyl alcohol and weathering) make it suitable for application in industrial or outdoor environments.

The film obtained after the coalescence and cross-linking phase has the following characteristics:

- High bonding properties on various substrates.
- Excellent mechanical properties and good abrasion resistance.
- Good resistance to solvents and ethyl alcohol.
- Excellent resistance to weathering.

## USES

Transparent finish for resin coatings. Transparent finish for decorative coatings and paints. Varnishing of concrete floors.

## SUBSTRATE PREPARATION

Every substrate must be clean, degreased and free from old coatings that are flaking off. When finishing resin coatings, the overcoating time of the previously applied product must be checked. Lightly abrade the surface to promote adhesion. It is advisable to use light-coloured abrasive papers with grit 150 or finer. After abrading, remove all dust from the surface completely.

## APPLICATION

At the time of application, add component B to component A, mixing carefully with a mechanical stirrer. Apply the mixture by roller with a consumption of 0,100-0,125 kg/m$^2$. For application, the use of suitable medium-pile rollers is recommended. To obtain a rough, non-slip surface, add 3% by weight of FINE GLASS BEADS to the mixture (see technical data). Avoid excessively thick films to allow the correct development of the nominal chemical-physical characteristics of the film. If necessary, the mixture can be diluted with water up to a maximum of 10%.

Product for professional use: the purchaser undertakes to strictly follow the warnings given above when applying the purchased product, as well as the instructions in the safety data sheet.

## PRODUCT DATA

| DESCRIPTION | RESULT |
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| Colour | Transparent |
| Consumption | By roller: 0,100-0,125 kg/m$^2$ |
| Specific gravity (at 25°C) | Mixture A+B: 1,00 ± 0,05 g/ml |
| Viscosity (at 25°C) | Mixture A+B: 420 ± 50 mPa·s (spindle 2, 30 rpm); mixture A+B diluted 10%: 210 ± 50 mPa·s (spindle 1, 12 rpm) |
| Solids content (A+B) | 38-40% by weight |
| VOC ready for use (Italian Legislative Decree 161/06) | < 70 g/l – Cat. A/j: two-component high-performance coating (BA – water-based) |
| Flash point | Not applicable |
| Tool cleaning | Water |
| Storage | 12 months, in a dry place at a temperature between 5°C and 35°C; pay particular attention to component B, which is sensitive to air humidity |

## APPLICATION DATA AND TIMES

| DESCRIPTION | RESULT |
|---|---|
| Mixing ratio (by weight) | A = 100, B = 22 |
| Recommended thinner | Water (only after mixing the 2 components), max 10% |
| Pot life (50% RH) | at 10°C: 120-180 min; at 25°C: 90-120 min; at 30°C: 60-90 min |
| Touch dry (50% RH) | at 10°C: 12-14 hours; at 25°C: 4-6 hours; at 30°C: 2-3 hours |
| Overcoating (50% RH) | at 25°C: from 4 to 36 hours |
| Full traffic (50% RH) | at 25°C: 3-5 days |
| Full curing (50% RH) | at 25°C: 7 days |
| Surface temperature | Between +10°C and +25°C |

| DESCRIPTION | RESULT |
|---|---|
| Ambient conditions for use | Temperature between +10°C and +30°C, RH < 70% |
| Maintenance of the coating | Wait at least 7 days after application of the coating before carrying out cleaning operations, and use only neutral detergents |

## TECHNICAL PERFORMANCE DATA

| DESCRIPTION | RESULT |
|---|---|
| Appearance | Slight orange-peel effect |
| Gloss at 60° | 70-80 (*) – glossy version |
| Slip resistance (Italian Ministerial Decree 236/89) | μ > 0,40 (with 3% FINE GLASS BEADS) |

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| CE marking (Reg. No. 305/2011) | Compliant with EN 1504-2. Coating (C), Principle (PR) |
| Pull-off adhesion (EN 1542) | 2,4 MPa |
| Abrasion resistance (EN ISO 5470-1) | 52 mg (Taber CS-10, 1000 cycles, 1000 g) |
| Impact resistance (EN ISO 6272-1) | > 9,8 N · m |
| Water permeability (EN 1062-3) | w < 0,002 kg/(m$^2$ · h^0,5) |

(*) The gloss level of the applied DS-POLTEC W is influenced by the porosity of the substrate and the mixing method.

## WARNINGS

Owing to the drying speed of this type of product, pay particular attention during application in order to avoid defects such as roller marks, overlaps and differences in gloss. In general, spread the varnish evenly over the substrate in the right quantity; when joining up with the area just varnished, resume varnishing a few centimetres away so as to discharge the excess product and then blend into the previous area, feathering and carefully dosing the quantity applied. Do not wait too long between overlaps: always varnish "wet-on-wet". Take care not to leave any unvarnished areas; if in doubt, the application of two coats of product is recommended. When covering large surfaces, the coordinated movement of several applicators is required.

The technical data given in this bulletin were obtained using standardised tests and standards, or modifications thereof, according to Design Systems systems. The methods applied may be requested from our technical service. The contents of this technical bulletin represent the best of our knowledge and experience. However, for uses other than those indicated, preliminary tests are recommended. In any case, Design Systems Srl accepts no responsibility for damage or defects caused by the use of its products, since the conditions of use are not under its control. It also advises that its technical service is available to customers for any information concerning the correct use of its products.